
DATA 780 Final Project

Jordan Bailey

University of North Carolina at Chapel Hill
jorbai@unc.edu

Scott Smith

University of North Carolina at Chapel Hill
smittsco@unc.edu

Chris Gonzales

University of North Carolina at Chapel Hill
chjgonz@unc.edu

1 Introduction

Recidivism refers to when a person who has previously been convicted of a crime reoffends—meaning they return to criminal behavior after being released from custody, completing a sentence, or undergoing rehabilitation. Researchers and policymakers often measure recidivism by re-arrest, reconviction, or re-incarceration rates within a specific period.

As data-driven tools are increasingly used to predict recidivism, concerns about fairness and bias have become critical. This is particularly significant because predictive models influence sentencing decisions, parole eligibility, and resource allocation for rehabilitation programs.

COMPAS (Correctional Offender Management Profiling for Alternative Sanctions) is a leading algorithmic tool for predicting recidivism, yet it has been widely criticized for machine bias and for failing to consider non-algorithmic factors such as socioeconomic conditions. Additionally, COMPAS does not account for systemic biases within the criminal justice system or disparities in housing, employment, and social support networks.

Disparate impact is often evaluated using measures such as the 80% rule, which assesses whether outcomes disproportionately affect protected groups, and the balanced error rate (BER), which tests whether protected attributes can be predicted from other features. The study seeks to develop a model that minimizes these disparities and reduces disparate impact while maintaining predictive accuracy.

2 Related Work

Our research focused on experiments where bias influenced the outcome an individual would face when being disproportionately impacted by protected classifiers. Outlined below is the research we have used as a platform for our experiment.

2.1 J. Dressel H. Farid

Dressel Farid argue that COMPAS is an expensive software that does no better than an individual with no experience in Criminal Justice could by themselves.

The nature of COMPAS and it's 14 features for predicting the probability that a criminal will recidivise upon release. The project was run on a 2013-2014 post-trial dataset using a classifier with a linear kernel and a non-linear SVM.

Their findings were that the algorithmic approach was no better at accurately classifying the data than a random subset of humans given a choice between two data points. It was further tested to include the race of the criminal, which led to biased and less accurate results.

2.2 S. Verma J. Rubin

Verma Rubin dive into the specificity of what makes something “fair” or not. They used the German Credit Rating dataset to study and quantify the “fairness” of credit ratings.

What they found were that the approach to credit is inherently biased. In fact, the causal model was found to be unfair. The classifier does a decent job classifying those with good credit, but in a head-to-head it was mildly harmful towards women.

The values for men and women balanced when training the model without sex data.

2.3 M. Feldman S.A. Friedler J. Moeller

Unintended bias is encoded via disparate impact, which is when there are vastly different outcomes given the same process across all groups.

Algorithms are showing this bias, and this paper implemented an 80

This is called a balanced error rate (BER), and if the algorithm can predict the protected attribute from the other features, then disparate impact is possible. They provide a model that does not show a disparate impact, and that their model lessens disparate impact.

2.4 S.A. Friedler C. Scheidegger S. Venkatasubramanian

This comparative study evaluated different fairness-enhancing interventions across multiple datasets, examining the tradeoffs between fairness and accuracy, and showing that no single approach dominates across all contexts.

2.5 F. Di Gennaro T. Laugel V. Grari X. Renard M. Detyniecki

This paper outlines a Post-Processing, Ratio-Based Debiasing Method, along with the math necessary to do so. The methodology and approach are directly related to our overall desired outcome, and we modeled our post-processing model after the work outlined in this paper.

3 Methods

We adopt and synthesize several techniques from our literature search to apply to our dataset. Ultimately we hope to arrive at a state-of-the-science solution for fairness testing.

3.1 Disparate Impact Test (80% Rule)

Feldman describes this legal definition from the US Equal Employment Opportunity Commission, which states that disparate impact occurs when the selection rate for a protected group is 80

$$\frac{P(C = \text{yes} \mid X = 0)}{P(C = \text{yes} \mid X = 1)} \leq \tau = 0.80 \quad (1)$$

where C indicates selection (or outcomes, more broadly), and X indicates a protected class status (race, sex, religion, etc.). In this case, $P(C=\text{yes}|X=0)$ is the portion of protected class individuals with positive outcomes. It is often applied in employment, lending, and socioeconomic metrics. We can use this definition as a first primary indicator for bias in a classifying algorithm.

3.2 Balanced Error Rate (BER) for Predictability Testing

Also from the Feldman paper, this is an error measure which distinguishes itself from accuracy. Typical accuracy metrics fail to measure the quality of a model when datasets are highly unbalanced. For example: consider a model meant to detect a rare condition. 1% of patients have the condition, and 99% are unaffected. A classifier that always predicts negative will have 99% accuracy, but is useless for learning about the condition.

Protected minority classes are analogous. In order to fairly consider them in your model, the error metric must take into account the “accuracy” for both classes equally. This is illustrated in the BER calculation. It is the average between the false negative rate for majority and the false positive rate for minority.

$$BER(f(Y), X) = \frac{\Pr(f(Y) = 0 \mid X = 1) + \Pr(f(Y) = 1 \mid X = 0)}{2} \quad (2)$$

The core innovation is how it measures how well the protected attribute X can be predicted from the other attributes Y . This makes it uniquely suited to detect disparate impacts.

Feldman then describes feeding the dataset into the so called certification algorithm. This is a classifier optimizing for (minimizing) BER, attempting to predict the protected attributes X from the remaining attributes Y . Consider: If data points can be correctly predicted to be minority class $X=0$, based on features Y , then those features correlate to that class. This is an indicator of bias and unfairness.

Ultimately, the maximum threshold for BER is derived as the ϵ -fairness, where $\epsilon = \frac{1}{2} - \frac{\beta}{8}$ and β is the fraction of the minority class selected.

This is one of the metrics we incorporate.

4 Exploratory Data Analysis

4.1 Initial Findings

Our Exploratory Data Analysis began with the distribution, structure, and discerning relevant characteristics of the COMPAS dataset necessary for analysis.

Our seven-feature analysis was comprised of:

- Age
- Priors Count
- Juvenile Felony Offense Counts
- Juvenile Misdemeanor Offense Counts
- Juvenile Miscellaneous Offense Counts
- Sex
- race

This included our two feature analysis features: Age & Priors Count.

The COMPAS distribution results show that African Americans have more offenders by a factor of roughly one and a half than Caucasians.

Table 1: Distribution of Records by Race in the COMPAS Dataset

Race	Count of Records
African-American	3,696
Asian	32
Caucasian	2,454
Hispanic	637
Native American	18
Other	377

Distribution of ages is heavily concentrated with most individuals being between 18 and 35 years of age. This concentration implies that age may be a strong predictor in modeling behavior.

Prior Offenses also shows a right-skew, with most records having 2 or less prior convictions. Very few individuals offend 5 or more times, leading to a sharp drop in records as prior convictions increase.

5 Preliminary Results

Our preliminary results match the findings of the original study into COMPAS, with our 7-feature Logistic Regression, 2-feature Logistic Regression, and Non-Linear SVM falling within the range of results for the entire study, without racial partitions.

Dressel and Farid (2018) highlight that simple models are as effective as comprehensive and proprietary models.

Our 7-feature Logistic Regression model had an accuracy of 0.676, and an AUC of 0.727 which is almost identical to COMPAS’ original findings. In fact, even the 2-feature (formatted for only age and prior offenses) only lost a little AUC in our experiment, yielding an accuracy of 0.672 with an AUC of 0.724, just below the 7 feature approach.

Our SVM didn’t outperform any of our models, or the findings from Dressel and Farid either. With an accuracy of 0.672, the SVM saw it’s AUC fall significantly, to 0.663.

Table 2: Performance comparison across models.

Model	Accuracy	AUC
7-feature Logistic Regression	0.676	0.727
2-feature Logistic Regression	0.672	0.724
7-feature Nonlinear SVM (RBF)	0.672	0.663

Table 3: Confusion matrix (7-feature Logistic Regression).

	Predicted: No	Predicted: Yes
Actual: No Recidivism	766	219
Actual: Recidivism	378	482

Table 4: Confusion matrix (2-feature Logistic Regression).

	Predicted: No	Predicted: Yes
Actual: No Recidivism	760	225
Actual: Recidivism	381	479

Table 5: Confusion matrix (7-feature Non-Linear SVM).

	Predicted: No	Predicted: Yes
Actual: No Recidivism	775	210
Actual: Recidivism	396	464

It’s clear across all three models that the frequency of False Positives is far more likely than the frequency of False Negatives.

Because over-prediction of risk can disproportionately affect certain groups, understanding where and why these errors happen is paramount.

While this initial run didn’t include racial partitions, our upcoming analysis will measure Balanced Error Rate (BER) and Disparate Impact to evaluate those specific fairness effects.

As a sensitivity check, we shuffled the protected attributes before we trained our model. Model performance didn’t change more than 0.01 for either accuracy or AUC. We can infer that group membership among the recidivists was not used for judgement. We aim to improve upon that in our final state.

6 Bias Evaluation Machine

6.1 Approach

The Bias Evaluation Machine (BEM) is a class that we created as the foundation for a future state library to test parity, bias, and fairness in training models across Data Science.

The principle is simple, by identifying a Sensitive Feature, we can perform a high volume of very small steps to routinely scale back the weight imparted by the sensitive variable.

Our inspiration came from the paper "Post-processing fairness with minimal changes" by F. Di Gennaro, T. Laugel, V. Grari, X. Renard, and M. Detyniecki. Their approach is simple, compare their Model Agnostic, Optimized Post-Processing Model against other known methods for Post-Processing.

Their results were model agnostic, did not require a sensitive label at the time of processing, and was built to minimize the number of prediction changes.

6.2 Parity Assessment

One of the most important parts of this approach comes from defining what their experiment defines as "Demographic Parity". We later reference this as Statistical Parity or "Gap".

$$P\text{-Rule} = \min \left(\frac{\mathbb{P}(\hat{Y} = 1 \mid S = 1)}{\mathbb{P}(\hat{Y} = 1 \mid S = 0)}, \frac{\mathbb{P}(\hat{Y} = 1 \mid S = 0)}{\mathbb{P}(\hat{Y} = 1 \mid S = 1)} \right). \quad (3)$$

6.3 Remapping Procedure

Following this, we set out to create a new mapping $g(x)$ which minimally adjusts the base predictions $f(x)$ of a trained model.

We compute a log loss as binary cross-entropy loss and build in methods to prevent overcorrection when adjusting the bias.

$$\begin{aligned} \min_g \mathbb{E}[\mathcal{L}_Y(\hat{Y}^g, Y)] \\ \text{s.t. } |\mathbb{E}[\hat{Y}^g \mid S = 1] - \mathbb{E}[\hat{Y}^g \mid S = 0]| \leq \varepsilon \quad & \text{(Fairness constraint)} \\ \mathbb{E}[\mathbf{1}_{\hat{Y}^g \neq \hat{Y}^f}] \leq \eta \quad & \text{(Minimal-change constraint)} \end{aligned} \quad (4)$$

The rescaled version $g(x)$ yields a score of who's prediction is more fair, and weighted by a factor of $r(x)$.

$$g(X) = \frac{1}{1 + \exp(-r(X) f_{\text{logit}}(X))}, \quad \text{where } r(X) = 1 + Xw \quad (5)$$

6.4 Ratio-Based Debiasing

We finalize our class with a Post-Processing method for Ratio-Based Debiasing, implementing a method which minimizes a composite loss function that jointly balances predictive accuracy, fairness, and stability of correction terms. The optimization objective can be expressed as:

$$\min_w \mathcal{L}_{\text{total}} = \mathcal{L}_Y(\hat{Y}^g, Y) + \alpha \mathcal{L}_{\text{fair}}(\hat{Y}^g, S) + \beta \mathcal{L}_{\text{ratio}}(r(X)), \quad (6)$$

where $\mathcal{L}_Y$ is the binary cross-entropy loss, $\mathcal{L}_{\text{fair}}$ measures the absolute prediction gap between sensitive groups,

$$\mathcal{L}_{\text{fair}} = |\mathbb{E}[\hat{Y}^g \mid S = 1] - \mathbb{E}[\hat{Y}^g \mid S = 0]|, \quad (7)$$

and $\mathcal{L}_{\text{ratio}}$ penalizes large corrective factors to maintain numerical stability,

$$\mathcal{L}_{\text{ratio}} = \mathbb{E}[(r(X) - 1)^2], \quad \text{with } r(X) = 1 + Xw. \quad (8)$$

The parameters α and β control the trade-off between fairness and overcorrection, respectively. We update the weights w iteratively via gradient descent:

$$w \leftarrow w - \eta \nabla_w \mathcal{L}_{\text{total}}, \quad (9)$$

where η is the learning rate. This approach serves as a simplified and fully model-agnostic post-processing debiasing method.

7 Findings

Our novel BiasEvaluationMachine class contains a disparate impact method, which summarizes the difference in positive outcomes using the DI ratio, and bootstrapped confidence interval method to calculate the confidence interval of our bias evaluation. We performed these tests on the base state of the models and the post-processing de-biased versions.

For the seven feature nonlinear support vector machine (NLSVM), the gap is shown to significantly decrease from 0.267 to 0.248. This means that difference between the predicted recidivism rates between Caucasian and African American records have become closer. In terms of disparate impact, for every one hundred African Americans recidivating, there are 68 Caucasian individuals recidivating, rather than 66. The 95% confidence interval for the DI ratio is 0.611 to 0.700.

The results are visualized in the accompanying Python notebook (see Appendix A). The post-processed de-biased model predicts a higher rate of recidivism for both classes, but a greater change for the rate of Caucasian records. This places the model closer to the center diagonal “line of equity,” representing statistical parity.

The BER of the NLSVM was 0.337, and the BER of the post-processing version of the model was 0.331. This indicates that the ratio-based de-biasing method does not make significant prediction changes, meaning disruptions to the accuracy or quality of the model, despite increasing fairness. The model does not push the predictions in a straight line towards the line of equity, rather it scales the logits with constraints to minimize changes and the sensitivity gap.

When comparing the raw seven feature linear regression model and the post-processed version, we did not find statistically significant differences in parity or ratio. Both versions of the model predict the same rates of recidivism. Thus, the BER also stayed at 0.331. There are several possible reasons for this. One is that the “change penalty” beta was set too high. This hyperparameter constrains the ratio function from straying too far from $r(x) = 1$, meaning it will not change the model. Further tuning of the hyperparameters would be needed to investigate this. Another possibility is that the seven feature regression model itself did not contain much demographic significance to start with. This seems unlikely given the imbalance, but if the ratio method does not detect fairness loss, then it will push the ratio towards 1.

Table 6: Fairness metrics for post-processed models.

Name	Outcome Gap	DI Ratio	CI
LR7	0.248	0.678	0.093
LR7_ber	0.248	0.678	0.093
NLSVM	0.270	0.655	0.090
NLSVM_ber	0.267	0.665	0.091

8 Future Steps

8.1 Library Development

Our future steps involve creating a library that can be used to judge bias and fairness within a model. Much like Pandas is the standard for dataframes, we would like Data Scientists, Analysts, and ML Enthusiasts to know there's a standard they can call on to evaluate, mitigate, and understand their model's bias.

By developing processes that can be run over schema agnostic models, we expand the tools and capabilities of Data Scientists, while also helping establish standard practices in assessing bias, and approaches to minimizing it.

8.2 ZafarAccuracy ZafarFairness

While this began as a facet of our project, we'd like to later implement this alongside functions like Disparate Impact.

Another attractive technique is described in the Friedler paper, which is a modification to typical classification algorithms like support vector machines. This distinguishes itself from the Disparate Impact Test, which is used to detect and reduce bias on training datasets. These techniques, then, are complimentary and can be combined into a more complete bias mitigation strategy.

To understand new forms of fairness constraints, we first understand that fairness is typically non-convex. That is, they arise from complex real-world relationships, with many local minima making solutions difficult. Optimizing large models for fairness is often time consuming and expensive.

Zafar describes a method to "relax" fairness constraints as convex parameters that have relatively simple solutions. This allows standard optimization techniques to be used. How do we arrive at such a simple universe?

Here, we modify the training algorithm by adding fairness constraints directly into the optimization objective of classifiers like logistic regression and SVMs. This works on the same principle as the Impact Test Rule (80% rule). The main idea here is the inclusion of decision boundary covariance, the covariance between the X (protected class, or classes) and the signed distance from Y (remaining) attributes to the decision boundary.

$$\text{Cov}(z, d_{\theta}(x)) \approx \frac{1}{N} \sum_{i=1}^N (z_i - \bar{z}) d_{\theta}(x_i) \quad (10)$$

Where z is the value of the protected class (X), $d_{\theta}(x)$ is the signed distance from the observation to the decision boundary, and θ are the classifier parameters. Consider that if the protected class z_i is zero for all observations, or the distance $d_{\theta}(x_i)$ is effectively zero for all observations, then this method reports a 100% fairness rate.

8.3 Github

Working Jupyter Notebooks and Resources are available using our Repo link below.

<https://github.com/uncjorbai/Data780FinalProject>

While base code is available, how to install, version history, and other helpful resources would be great additions in future states of this project.

One day, we hope to have this under a more permanent name and identity.

9 Conclusion

Through Data Cleaning and Exploratory Data Analysis we determine bias according to distribution of our sensitive variable when evaluating COMPAS data for race, age, and priors.

Using our preliminary findings, we set out to build, replicate, and train three distinct classification models for determining bias in Machine Learning. Through our seven-feature and two-feature logistic regression models and our seven-feature non-linear support vector machine, we determined that Caucasian individuals were predicted less likely to recidivise in 2 years than their African-American counterparts.

Using the COMPAS dataset, we demonstrated how standard predictive models — 7-feature and 2-feature logistic regression as well as a non-linear SVM — can be embedded inside a dedicated fairness-evaluation framework rather than treated as black boxes. We implemented a custom Bias Evaluation Machine class as a wrapper for trained data, computes bias and fairness statistics on split data, and supports post-processing via a ratio-based debiasing method that uses a simple gradient approximation to adjust predictions with respect to a sensitive variable. This combination of model-agnostic evaluation, debiasing, and detailed logging of loss and fairness gaps provides a reusable template that others can adapt when auditing high-stakes models for bias, extending the typical “fit-predict-accuracy” workflow into a more responsible, fairness-aware pipeline.

Alongside Balanced Error Rate and bootstrapped CI’s to determine our fairness, we have created the foundations of a library useful for Data Scientists in any field who need to determine how certain fields will impact the distribution of their findings and predictions. We learned that building a “good” model requires more than strong accuracy scores—especially with sensitive datasets like COMPAS. Our EDA showed clear structural imbalances in race, age, and priors count, and these patterns helped us anticipate where bias could emerge before any modeling began.

By evaluating multiple models we saw that accuracy alone masked meaningful differences in outcomes across racial groups. Using fairness diagnostics such as Disparate Impact ratios, BER, and bootstrapped confidence intervals, we learned how to quantify and compare those disparities in a statistically meaningful way.

We also learned how to move beyond measuring bias by implementing our own Bias Evaluation Machine (BEM). Working through the BEM’s gradient-based adjustment process showed us how fairness gaps change over iterations, and it clarified the trade-offs between fairness and model confidence. This taught us that fairness isn’t a single metric—it requires balancing multiple objectives and understanding how even small adjustments can affect group-level outcomes.

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10 Appendix / supplemental material

A Accompanying Notebook

The Jupyter notebook used for generating plots, computing fairness metrics, and visualizing post-processed results can be found at:

https://github.com/uncjorbai/Data780FinalProject/blob/main/DATA780_Final.ipynb

This notebook includes the full workbook accompanying this experiment.